

Talking LED Alarm Clock Using TM1637 LED Module

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Every microcontroller project needs a display. An LCD display with one or two rows for characters is a good choice but it needs several wire connections to make it work. There are seven segment LEDs which are either common-anode or common-cathode type. Earlier, these LEDs were used for number displays which had their pros and cons.

Suppose a 4-digit clock display is required, it needs four separate LEDs. These are wired in common for segments (cathodes) and the anode of each LED is separately excited, one after another, with a dwell time for each digit display of a few milliseconds in a multiplexed manner. Because of persistence of vision, the four LEDs appear as a continuous display, though they burn one after another.

Most projects use this method only for four or more digits for display. With eight wires to connect to segments (seven for the segments and one for decimal point) and four multiplexing anode signals to provide power supply one by one, totally $8 + 4 + 1 = 13$ wire connections are needed for a 4-digit clock's speed display. The last '1' is for the common ground wire.

With the TM1637 LED module (Fig. 1) becoming easily available at an affordable price of around ₹90, projects of this kind have become extremely simple to construct. Using an Arduino Uno board, or even with a plain microcontroller chip, these projects are simple, quick, and easy to fabricate within limited cost. The simplicity lies in the fact that the TM1637 module comes with just four pins for connections. Out of the four, two are 5V and ground connections for power supply and the other two are for a two-wire (clock and data) interface like the so-called I2C bus which one can learn using the TM1637.



Fig. 1: TM1637 LED module

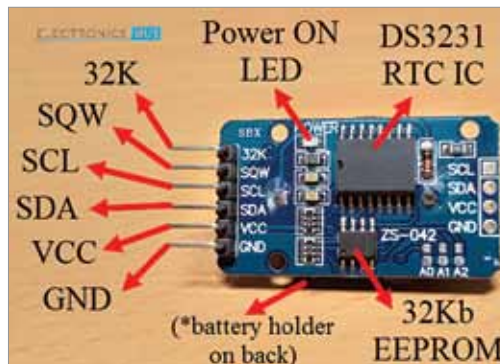


Fig. 2: DS3231 RTC module

Illustrative example

The TM1637 module's data and clock pins are interfaced to the Arduino board through pins 8 and 7, respectively. Other than these two wires, 5V and ground wires are also connected to Arduino. The TM1637 module displays time with a central blinking colon.

To get the real time, an easily-available DS3231 real-time clock (RTC) module is interfaced to the Arduino board. When switching on the circuit, the time shown is correct time that need not be corrected at all (except one time initial settings in the source code).

The RTC module also has just two wire connections. SCL and SDA pins of the RTC module are connected to A5 and A4 pins of Arduino board,

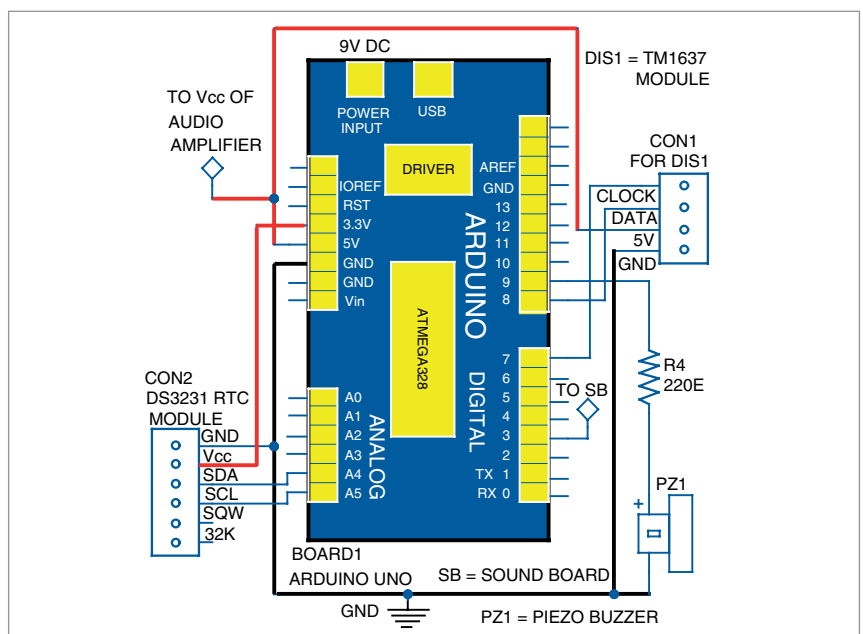


Fig. 3: Circuit diagram of the talking LED alarm clock with Arduino